Instructor Key: Day 1 Mastery Quiz v2

Visible Output and Stored State

Separate key for `DAY01_VISIBLE_OUTPUT_EVIDENCE_QUIZ_STUDENT_06192026_v2.pdf`. Harder original ramp-gate record-check scenario.

1 Artifact status, source boundary, and difficulty bump

Field	Status
Student artifact	Harder 10-point Day 1 mastery quiz; quiz, not workbooklet.
Target	Separate stored state from printed output and notebook display; trace assignment order; distinguish displayed candidate, printed stored value, separately stored report value, and later candidate update.
Difficulty bump from v1	v2 removes one trace row, adds assignment-order reasoning in Cell D, and adds a second tempting-but-wrong record claim. Students must notice that `reported_speed` changing after `speed_mps = reported_speed` does not retroactively change `speed_mps`.
Scenario/source boundary	Original ramp-gate record-check scenario. No protected exam/review prompt text, answer-key language, private student data, or near-copy source structure used.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.

2 Reference code

```python
# Cell A: first gate check
run_id = "gate-1"
distance_m = 10.0
time_s = 4.0
speed_mps = distance_m / time_s
print("raw", speed_mps)
speed_mps

# Cell B: second gate check with a candidate correction
run_id = "gate-2"
time_s = 2.0
speed_mps = distance_m / time_s
correction_mps = 0.5
speed_mps + correction_mps
print("stored after candidate", speed_mps)

# Cell C: store the reviewed report value separately
reported_speed = speed_mps + correction_mps
print("reported", reported_speed)

# Cell D: record, then make one more candidate update
speed_mps = reported_speed
reported_speed = reported_speed + 0.2
print("final stored", speed_mps)
print("latest candidate", reported_speed)
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Common issue / support move	Tags
1	Physical measurement inputs	`distance_m` and `time_s`. `run_id` is a label, `speed_mps` and `reported_speed` are calculated/corrected values.	If students list every variable, ask which values came from the ramp-gate measurement.	recognize; explain
1	First display-only `5.5`	`speed_mps + correction_mps` in Cell B. It can display `5.5` as a final expression but does not store that value in either named speed variable.	Ask where the equals sign is and whether `reported_speed` exists yet.	recognize; predict
1	After Cell B trace	`speed_mps = 5.0`; `reported_speed = not defined`; visible evidence includes displayed `5.5` and printed `stored after candidate 5.0`.	Strong misconception: display-only `5.5` becomes stored. Rerun assignment-vs-expression mini example.	trace; predict
1	After Cell C trace	`speed_mps = 5.0`; `reported_speed = 5.5`; visible evidence prints `reported 5.5`.	Students may overwrite `speed_mps` too early. Ask which variable is left of the equals sign.	trace; explain
1	After Cell D trace	`speed_mps = 5.5`; `reported_speed = 5.7`; visible evidence prints `final stored 5.5` and `latest candidate 5.7`.	Hardest v2 misconception: later changing `reported_speed` also changes `speed_mps`. Emphasize assignment order and numeric copy.	trace; explain
1	Claim 1 debug	Wrong/incomplete: after Cell B, `5.5` was visible as a display-only candidate. Stored `speed_mps` was still `5.0`; `reported_speed` was not defined.	Require displayed/stored language rather than just "not assigned."	debug; explain
1	Claim 2 debug	Wrong: Cell D first copies current `reported_speed` into `speed_mps`, so `speed_mps` becomes `5.5`. Then `reported_speed` becomes `5.7`; that later update does not change `speed_mps`.	If students answer `5.7`, have them annotate the two Cell D assignments in order.	debug; trace
1	Non-mutating check	Accept `print(speed_mps)` or a final bare expression `speed_mps`. Do not accept any assignment.	If student writes `speed_mps = ...`, mark as changing state, not checking it.	test; debug
1	Safer edit	`reported_speed = speed_mps + correction_mps`. Accept missing variable name `reported_speed`.	If student uses `speed_mps` on the left, they overwrite too soon.	implement; trace
1	Transfer/evidence reason	Secure response: the two Cell D prints show the recorded stored value and the later candidate separately; printing only the latest candidate would hide that `speed_mps` remains `5.5`.	If generic, ask which value would go in the lab table and which is only the next candidate.	transfer; explain

4 Expected trace table

Checkpoint	Stored `speed_mps`	Stored `reported_speed`	Visible evidence that could mislead a record
After Cell B	`5.0`	not defined	Displayed `5.5`; printed `stored after candidate 5.0`.
After Cell C	`5.0`	`5.5`	Printed `reported 5.5`; no overwrite of `speed_mps`.
After Cell D	`5.5`	`5.7`	Printed `final stored 5.5`; `latest candidate 5.7`.

5 Mastery interpretation and support routing

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10, with trace/debug evidence present	Student can separate display, print, assignment, separate storage, and later assignment-order effects.	Ready for Day 2 state/evidence warm-up; include one cumulative trace row.
Developing	5–7 / 10, or uneven tag evidence	Student calculates most values but confuses displayed candidate with stored value or misses the Cell D assignment order.	Target trace/debug tags; rerun a two-variable copy/update mini-trace.
Needs support	0–4 / 10, or no usable trace/debug evidence	Student cannot yet identify which variable changes on each assignment line.	Model a smaller assign/print/bare-expression/overwrite example; complete first trace row together.

6 Optional recovery mini-surface

Use only if several students answer Cell D as if both variables end at `5.7`.

```
reading = 5.0
reported = reading + 0.5
reading = reported
reported = reported + 0.2
print(reading)
print(reported)
```

Ask students to write what each variable stores after each assignment. Secure recovery answer: `reading` ends at `5.5`; `reported` ends at `5.7`. The last assignment changes only `reported`.

7 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, print-safe margins, render sheets, answer-key separation, and absence of protected/private markers. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores. Working release threshold remains Ally at least 95% and Panorama at least 95%.